

Introduction to Markov Chain Monte Carlo

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1 Introduction to Bayesian Computation

In Bayesian inference, our goal is to compute posterior summaries of unknown parameters. Let θ denote the unknown parameter and let $\pi(\theta)$ denote its posterior density given the observed data. Typical quantities of interest include posterior means, posterior variances, posterior probabilities, credible intervals and posterior predictive distributions.

Many of these quantities can be expressed as expectations with respect to the posterior distribution. For example, if g is any measurable function, then our goal is to compute

$$E_{\pi}[g(\theta)] = \int g(\theta)\pi(\theta) d\theta. \quad (1)$$

Unfortunately, posterior distributions are often analytically intractable. In most interesting Bayesian models, the posterior density is only known up to a normalizing constant. Specifically, Bayes' theorem gives

$$\pi(\theta) = \frac{L(\theta)p(\theta)}{\int L(\vartheta)p(\vartheta) d\vartheta}, \quad (2)$$

where $L(\theta)$ denotes the likelihood and $p(\theta)$ denotes the prior density. The denominator

$$\int L(\vartheta)p(\vartheta) d\vartheta \quad (3)$$

is called the *marginal likelihood* or *evidence*. In many applications this integral is impossible to evaluate analytically and is computationally prohibitive even numerically.

Notice, however, that many Bayesian procedures require only the posterior distribution itself and not the value of the normalizing constant. This observation forms the basis of modern Bayesian computation.

Broadly speaking, there are two major classes of computational methods.

1. Markov Chain Monte Carlo (MCMC).
2. Variational Inference (VI).

Markov Chain Monte Carlo constructs a Markov chain whose stationary distribution is the posterior distribution. By simulating the chain for a sufficiently long time, one obtains approximate samples from the posterior.

Variational inference takes a completely different approach. Instead of simulation, it approximates the posterior distribution by a simpler family of distributions chosen by solving an optimization problem. Variational inference is usually much faster than MCMC but often introduces additional approximation error.

In these notes we begin with Markov Chain Monte Carlo.

2 The MCMC Principle

Suppose that our target distribution is a probability distribution π on a state space S . Throughout these notes, π will usually denote the posterior distribution of the unknown parameter.

If we could generate independent random variables

$$\theta^{(1)}, \dots, \theta^{(N)} \stackrel{\text{i.i.d.}}{\sim} \pi,$$

then by the ordinary law of large numbers,

$$\frac{1}{N} \sum_{t=1}^N g(\theta^{(t)}) \longrightarrow \int g(\theta) \pi(\theta) d\theta \quad (4)$$

almost surely.

The difficulty, of course, is that independent sampling from π is typically impossible.

The key idea behind MCMC is surprisingly simple. Instead of generating independent samples, we generate a sequence

$$\theta^{(0)}, \theta^{(1)}, \theta^{(2)}, \dots$$

whose successive values are dependent. The dependence is carefully constructed so that the sequence forms a Markov chain having stationary distribution π .

Although the samples are correlated, a remarkable theorem states that time averages still converge to posterior expectations. Thus, if

$$\theta^{(0)}, \theta^{(1)}, \dots, \theta^{(N)}$$

are generated by an appropriate Markov chain, then

$$\frac{1}{N} \sum_{t=1}^N g(\theta^{(t)}) \approx \int g(\theta) \pi(\theta) d\theta. \quad (5)$$

Equation (5) is the fundamental principle underlying all Markov Chain Monte Carlo methods.

The remarkable feature is that we never need to compute the normalizing constant of the posterior density. Most MCMC algorithms require only quantities proportional to $\pi(\theta)$, which is precisely what Bayes' theorem usually provides.

The remainder of this chapter is devoted to answering two questions.

1. How do we construct a Markov chain whose stationary distribution is π ?
2. Why do averages computed from the Markov chain converge to expectations under π ?

To answer these questions we first review the basic concepts of Markov chains.

3 Markov Chains

A sequence of random variables

$$\Theta^{(0)}, \Theta^{(1)}, \Theta^{(2)}, \dots$$

taking values in a state space S is called a Markov chain if, conditional on the present state, the future is independent of the past. More precisely,

$$P\left(\Theta^{(t+1)} \in A \mid \Theta^{(t)}, \Theta^{(t-1)}, \dots, \Theta^{(0)}\right) = P\left(\Theta^{(t+1)} \in A \mid \Theta^{(t)}\right) \quad (6)$$

for every measurable set A .

Thus the current state contains all the information required to generate the next state.

Throughout these notes we shall only consider *time-homogeneous* Markov chains, meaning that the transition rule does not depend on time.

If the state space is finite or countable, the Markov chain is completely specified by the transition probabilities

$$P(x, y) = P\left(\Theta^{(t+1)} = y \mid \Theta^{(t)} = x\right). \quad (7)$$

When the state space is continuous, $P(x, y)$ denotes the corresponding transition density, so that

$$P\left(\Theta^{(t+1)} \in A \mid \Theta^{(t)} = x\right) = \int_A P(x, y) dy. \quad (8)$$

In either case,

$$P(x, y) \geq 0, \quad (9)$$

and

$$\sum_y P(x, y) = 1 \quad (10)$$

in the discrete case, or

$$\int P(x, y) dy = 1 \quad (11)$$

in the continuous case.

The transition kernel P completely determines the evolution of the Markov chain.

3.1 Stationary Distributions

Suppose that $\{\Theta^{(t)}\}$ is a time-homogeneous Markov chain with transition kernel P .

A probability distribution π on the state space S is called a *stationary distribution* (or *invariant distribution*) if starting the Markov chain from π leaves the distribution unchanged after one step. In other words,

$$\Theta^{(t)} \sim \pi \implies \Theta^{(t+1)} \sim \pi.$$

If the state space is finite or countable, this condition is equivalent to

$$\pi(y) = \sum_{x \in S} \pi(x) P(x, y), \quad (12)$$

for every state $y \in S$.

When the state space is continuous,

$$\pi(y) = \int_S \pi(x)P(x, y) dx. \quad (13)$$

The stationary distribution is the equilibrium distribution of the Markov chain. If the chain is started from π , then every iterate has exactly the same distribution:

$$\Theta^{(0)}, \Theta^{(1)}, \Theta^{(2)}, \dots \sim \pi.$$

The central idea behind Markov Chain Monte Carlo is therefore to construct a Markov chain whose stationary distribution is the target distribution of interest.

3.2 Ergodic Theorem

Having a stationary distribution is useful only if the Markov chain actually converges to it. Fortunately, under mild regularity conditions this is exactly what happens.

A Markov chain is called *irreducible* if every state can be reached from every other state with positive probability after sufficiently many transitions.

For finite state spaces, irreducibility is often very easy to verify. For continuous state spaces, the precise definition is somewhat more technical, and we shall not pursue it here.

The following theorem is one of the fundamental results in the theory of Markov chains.

Theorem 3.1 (Ergodic Theorem). *Suppose that $\{\Theta^{(t)}\}$ is an irreducible Markov chain having stationary distribution π . Then, for every integrable function g ,*

$$\frac{1}{N} \sum_{t=1}^N g(\Theta^{(t)}) \longrightarrow \int g(\theta)\pi(\theta) d\theta \quad (14)$$

almost surely.

Thus, although the successive samples are dependent, averages along the trajectory converge to expectations under the stationary distribution.

This theorem plays the same role for Markov chains as the ordinary law of large numbers plays for independent random variables.

For MCMC, this theorem has an important practical implication. Once we have constructed a Markov chain whose stationary distribution is the posterior distribution, posterior expectations can be approximated by simple empirical averages computed along a single realization of the chain.

4 An Illustrative Example

Before developing general MCMC algorithms, let us look at a simple example that illustrates the main ideas.

Suppose we wish to generate uniformly from the collection of all binary sequences of length d having no two consecutive ones.

For example, when $d = 5$, the sequences

$$00000, \quad 01010, \quad 10100$$

are allowed, whereas

$$11000, \quad 01110$$

are not.

Formally, define the state space

$$S = \left\{ x = (x_1, \dots, x_d) \in \{0, 1\}^d : x_i x_{i+1} = 0 \text{ for every } i = 1, \dots, d-1 \right\}.$$

Our goal is to generate a sequence

$$X^{(1)}, X^{(2)}, \dots$$

whose limiting distribution is the uniform distribution on S .

At first sight this may appear to be a simple problem. One possible approach is rejection sampling: repeatedly generate an independent binary sequence and reject it whenever two consecutive ones appear.

Unfortunately, this method quickly becomes impractical as the length d increases.

Indeed, let N_d denote the number of binary sequences of length d having no consecutive ones. It can be shown that

$$N_d = F_{d+2},$$

where F_k denotes the k th Fibonacci number.

Since

$$F_d \sim \frac{\varphi^d}{\sqrt{5}},$$

where

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.618,$$

the acceptance probability of rejection sampling is

$$\frac{N_d}{2^d} \approx \left(\frac{\varphi}{2} \right)^d.$$

Because

$$\frac{\varphi}{2} \approx 0.809,$$

the acceptance probability decreases exponentially fast with the dimension.

For example,

$$d = 100$$

already gives an acceptance probability of approximately

$$0.809^{100} \approx 7 \times 10^{-10},$$

making rejection sampling essentially useless.

We therefore need a different strategy.

4.1 A Markov Chain on the State Space

The rejection sampler fails because it attempts to generate an entire binary sequence from scratch. Instead, let us proceed in a much more local manner. Suppose we already have one valid binary sequence. We shall make a small random modification to it while ensuring that the new sequence also belongs to the state space.

Suppose the current state of the Markov chain is

$$X = (X_1, \dots, X_d) \in S.$$

The Markov chain evolves as follows.

1. Choose one of the coordinates

$$i \in \{1, \dots, d\}$$

uniformly at random.

2. Replace the i th bit by its opposite value:

$$X_i \longrightarrow 1 - X_i.$$

3. If the resulting sequence still belongs to S , accept the proposed move. Otherwise reject it and remain at the current state.

Notice that every transition changes at most one coordinate of the binary sequence. Consequently, successive states of the Markov chain are very similar to one another. Such local moves are typical in Markov Chain Monte Carlo algorithms.

For example, suppose

$$X = (1, 0, 0, 1, 0).$$

If the second coordinate is selected, the proposal becomes

$$(1, 1, 0, 1, 0),$$

which contains two consecutive ones. This proposal is therefore rejected.

If instead the third coordinate is selected, the proposal becomes

$$(1, 0, 1, 1, 0),$$

which again violates the constraint and is rejected.

Finally, if the fifth coordinate is selected, the proposal becomes

$$(1, 0, 0, 1, 1),$$

which is also rejected.

On the other hand, if the first coordinate is selected, the proposal is

$$(0, 0, 0, 1, 0),$$

which is valid and is therefore accepted.

Thus some proposed moves are accepted while others are rejected, depending on whether the constraint is preserved.

4.2 Why Does This Work?

The Markov chain above was constructed almost by common sense. We only changed one coordinate at a time while preserving the constraint. Nevertheless, there are several nontrivial questions that naturally arise.

1. Can every valid binary sequence eventually be reached?
2. Does the chain become trapped inside some subset of the state space?
3. If we run the chain for a very long time, what distribution does it converge to?

The answers to these questions are provided by the general theory of Markov chains.

We first show that every state can be reached from every other state.

Proposition 4.1. *The Markov chain described above is irreducible.*

Proof. Take any two binary sequences

$$x, y \in S.$$

We shall show that there exists a sequence of valid moves connecting x to y .

Starting from x , first flip every coordinate equal to one. Since changing a one into a zero can never create two consecutive ones, each intermediate sequence remains inside S . After finitely many steps we arrive at the all-zero sequence

$$(0, \dots, 0).$$

Starting from the all-zero sequence, now flip precisely those coordinates where y has value one.

Because y itself belongs to S , no two adjacent coordinates are equal to one. Therefore every intermediate sequence obtained during this construction also belongs to S .

Hence there exists a path of positive probability from x to y . Since x and y were arbitrary, the chain is irreducible. $\square$

The next property is even simpler.

Proposition 4.2. *The Markov chain is aperiodic.*

Proof. Whenever a proposed move violates the constraint, the proposal is rejected and the chain remains at its current state.

Hence every state has a positive probability of remaining unchanged in one step. Therefore every state has period one, implying that the chain is aperiodic. $\square$

Thus the Markov chain satisfies the two basic conditions required for convergence: it is irreducible and aperiodic.

The only remaining question is to determine its stationary distribution.

Remarkably, this turns out to be extremely easy.

4.3 Symmetry and the Uniform Distribution

Recall that our goal is to sample uniformly from the state space

$$S = \left\{ x \in \{0, 1\}^d : x_i x_{i+1} = 0 \text{ for all } i \right\}.$$

The Markov chain constructed above proposes to flip one randomly chosen coordinate and accepts the proposal whenever the resulting sequence still belongs to S .

The key observation is that the transition mechanism is perfectly symmetric.

Indeed, suppose that two distinct states x and y communicate in one step. Then x and y differ in exactly one coordinate. The probability of moving from x to y is simply the probability of choosing that coordinate, namely

$$P(x, y) = \frac{1}{d}.$$

Conversely, starting from y , flipping the same coordinate takes us back to x . Hence

$$P(y, x) = \frac{1}{d}.$$

Therefore

$$P(x, y) = P(y, x) \tag{15}$$

for every pair of distinct states.

Equation (15) says that whenever the chain can move from one state to another, the reverse move is equally likely.

This immediately suggests that the uniform distribution should be the stationary distribution.

To prove this, let

$$\pi(x) = \frac{1}{|S|}$$

for every state $x \in S$.

Then

$$\pi(x)P(x, y) = \frac{1}{|S|}P(x, y) = \frac{1}{|S|}P(y, x) = \pi(y)P(y, x).$$

Thus the probability flowing from x to y is exactly equal to the probability flowing from y back to x .

This balance holds simultaneously for every pair of states.

Consequently, the total probability entering any state is exactly equal to the total probability leaving that state. Therefore the uniform distribution is unchanged after one step of the Markov chain.

Since the chain is irreducible and aperiodic, the ergodic theorem now implies that

$$\frac{1}{N} \sum_{t=1}^N g(X^{(t)}) \longrightarrow \frac{1}{|S|} \sum_{x \in S} g(x)$$

almost surely for every function g .

In particular, after running the Markov chain for a sufficiently long time, the current state is approximately uniformly distributed over all binary sequences having no consecutive ones.

4.4 A New Perspective

The example above illustrates the central philosophy of Markov Chain Monte Carlo.

Suppose our goal is to sample from some complicated probability distribution π .

Instead of attempting to sample directly from π , we proceed in two steps.

1. Construct a Markov chain whose state space is the same as that of the target distribution.
2. Design the transition probabilities so that π becomes the stationary distribution of the chain.

Once this has been achieved, the ergodic theorem guarantees that averages computed along the trajectory of the Markov chain converge to expectations under π .

The only remaining challenge is therefore the following.

Given an arbitrary probability distribution π , how can we construct a Markov chain having π as its stationary distribution?

The binary sequence example was relatively easy because the transition kernel happened to be symmetric. In most practical Bayesian problems, however, such symmetry is absent.

We therefore need a general principle for constructing Markov chains with a prescribed stationary distribution.

This principle is known as the *detailed balance condition*.

5 Detailed Balance

The binary sequence example illustrates an important general principle. The stationary distribution arose because, for every pair of states x and y , the amount of probability flowing from x to y was exactly equal to the amount flowing from y back to x .

This idea is known as *detailed balance*.

Suppose that π is a probability distribution on the state space S , and suppose that the Markov chain has transition kernel $P(x, y)$.

Imagine that at some instant the chain has already reached its equilibrium distribution π . The fraction of particles located at state x is then $\pi(x)$.

During one transition of the chain, a fraction

$$\pi(x)P(x, y)$$

of all particles move from x to y .

Similarly,

$$\pi(y)P(y, x)$$

is the fraction moving from y to x .

If these two quantities are equal, then there is no net probability flow between the two states.

This motivates the following definition.

Definition 5.1. *A Markov chain is said to satisfy the detailed balance condition with respect to a probability distribution π if*

$$\pi(x)P(x, y) = \pi(y)P(y, x) \tag{16}$$

for every pair of states $x, y \in S$.

Detailed balance is often called the *local balance condition* because it balances the probability flow between every pair of neighbouring states individually.

The importance of detailed balance comes from the following simple result.

Theorem 5.2. *If a Markov chain satisfies detailed balance with respect to a probability distribution π , then π is a stationary distribution of the chain.*

Proof. For every state y ,

$$\begin{aligned} \sum_{x \in S} \pi(x)P(x, y) &= \sum_{x \in S} \pi(y)P(y, x) \\ &= \pi(y) \sum_{x \in S} P(y, x) \\ &= \pi(y), \end{aligned} \tag{17}$$

because

$$\sum_{x \in S} P(y, x) = 1.$$

Thus

$$\pi(y) = \sum_{x \in S} \pi(x) P(x, y),$$

which is precisely the definition of a stationary distribution. $\square$

Notice that detailed balance is only a sufficient condition. A Markov chain may possess a stationary distribution without satisfying detailed balance. Nevertheless, detailed balance is extremely useful because it is easy to verify and leads to elegant constructions of Markov chains.

5.1 A Graphical Interpretation

Detailed balance has a simple physical interpretation.

Imagine that every state is represented by a vertex of a graph and the Markov chain moves along the edges of this graph.

Suppose the chain has already reached equilibrium.

For every edge joining two states x and y , detailed balance says that the amount of probability flowing from x to y is exactly equal to the amount flowing from y back to x .

Consequently there is no net probability current anywhere in the graph.

One may think of probability as an incompressible fluid flowing along the edges of the graph. At equilibrium, every pipe carries exactly the same amount of fluid in both directions. Since there is no net accumulation or depletion of probability at any vertex, the distribution remains unchanged after one transition.

This viewpoint provides perhaps the simplest intuition behind detailed balance.

5.2 An Example

Consider a random walk on an undirected graph.

Suppose the graph has vertex set

$$S = \{1, \dots, n\},$$

and let

$$d_i$$

denote the degree of vertex i .

The transition probabilities are

$$P(i, j) = \frac{1}{d_i} I\{i \sim j\},$$

where $i \sim j$ indicates that the two vertices are adjacent.

Now define

$$\pi(i) = \frac{d_i}{\sum_{k=1}^n d_k}.$$

Then

$$\begin{aligned} \pi(i)P(i, j) &= \frac{d_i}{\sum_k d_k} \cdot \frac{I\{i \sim j\}}{d_i} \\ &= \frac{I\{i \sim j\}}{\sum_k d_k}. \end{aligned} \tag{18}$$

Similarly,

$$\pi(j)P(j, i) = \frac{I\{i \sim j\}}{\sum_k d_k}. \tag{19}$$

Hence

$$\pi(i)P(i, j) = \pi(j)P(j, i),$$

so detailed balance holds.

Therefore the stationary distribution of the random walk is proportional to the vertex degrees.

Vertices having many neighbours are visited more frequently than vertices having few neighbours.

6 Towards Metropolis–Hastings

The examples we have seen so far were relatively easy because the transition probabilities were designed specifically so that detailed balance held.

In Bayesian statistics, however, the situation is quite different.

The target distribution

$$\pi(\theta)$$

is fixed in advance—it is the posterior distribution.

Our task is therefore exactly the reverse.

We already know the stationary distribution that we want.

The problem is to construct a Markov chain having this prescribed stationary distribution.

The Metropolis–Hastings algorithm provides a remarkably general solution to this problem.

The basic idea is extremely simple.

First construct any Markov chain that proposes candidate moves.

Then modify the chain by occasionally rejecting proposed moves so that the resulting chain satisfies detailed balance with respect to the desired target distribution.

The next section explains how this acceptance probability is derived.

7 The Metropolis–Hastings Algorithm

Suppose our target distribution is a probability distribution π on a state space S . We would like to construct a Markov chain whose stationary distribution is π .

A natural first attempt is to construct a Markov chain that proposes small moves.

More precisely, suppose that when the current state is x , we first generate a proposal

$$Y \sim Q(x, \cdot),$$

where $Q(x, \cdot)$ is some proposal distribution chosen by us.

Unfortunately, there is no reason why the Markov chain generated by Q should have stationary distribution π . In general,

$$\pi(x)Q(x, y) \neq \pi(y)Q(y, x),$$

so detailed balance fails.

The key idea of Metropolis and Hastings is remarkably simple.

Instead of accepting every proposed move, we accept only a suitable fraction of them. By rejecting some proposed moves, we reduce the probability flow in whichever direction it is too large until detailed balance holds.

7.1 Deriving the Acceptance Probability

Suppose that when the proposal chain suggests a move

$$x \longrightarrow y,$$

we accept this move with probability

$$A(x, y).$$

If the proposal is rejected, the chain simply remains at the current state.

The resulting transition probabilities are therefore

$$P(x, y) = Q(x, y)A(x, y), \quad x \neq y. \quad (20)$$

Our goal is to choose $A(x, y)$ so that detailed balance holds.

Substituting (20) into the detailed balance condition gives

$$\pi(x)Q(x, y)A(x, y) = \pi(y)Q(y, x)A(y, x). \quad (21)$$

Thus the acceptance probabilities must compensate exactly for the difference between the forward and backward probability flows.

Suppose first that

$$\pi(x)Q(x, y) > \pi(y)Q(y, x).$$

In this case there is too much probability flowing from x to y .

The simplest correction is to accept every move from y back to x , that is,

$$A(y, x) = 1,$$

and reduce only the flow from x to y .

Equation (21) then gives

$$A(x, y) = \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)}.$$

Notice that this quantity is automatically less than one.

Now suppose instead that

$$\pi(x)Q(x, y) < \pi(y)Q(y, x).$$

Now the excess flow is in the opposite direction.

By exactly the same reasoning, we accept every move from x to y ,

$$A(x, y) = 1,$$

and reduce only the reverse flow.

Combining the two cases yields the remarkably simple rule

$$A(x, y) = \min \left\{ 1, \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)} \right\}. \quad (22)$$

Thus the acceptance probability is not mysterious at all. It is simply the smallest modification of the proposal chain that restores detailed balance.

7.2 The Metropolis–Hastings Algorithm

The Metropolis–Hastings algorithm may now be summarized as follows.

1. Initialize at an arbitrary state

$$\theta^{(0)}.$$

2. For $t = 0, 1, \dots$

- a) Let

$$x = \theta^{(t)}.$$

- b) Generate a proposal

$$y \sim Q(x, \cdot).$$

- c) Compute

$$r = \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)}.$$

- d) Generate

$$U \sim \text{Uniform}(0, 1).$$

- e) If

$$U \leq \min(1, r),$$

set

$$\theta^{(t+1)} = y.$$

Otherwise set

$$\theta^{(t+1)} = x.$$

Observe that only the ratio

$$\frac{\pi(y)}{\pi(x)}$$

appears in the algorithm.

Consequently, the normalizing constant of π cancels.

This is one of the principal reasons why Markov Chain Monte Carlo is so powerful in Bayesian statistics. We only need to know the posterior distribution up to proportionality.

7.3 Why the Algorithm Works

The Metropolis–Hastings algorithm is designed so that the resulting Markov chain satisfies detailed balance.

Theorem 7.1. *The Metropolis–Hastings Markov chain has stationary distribution π .*

Proof. Suppose first that

$$x \neq y.$$

The transition probability from x to y is

$$P(x, y) = Q(x, y) \min \left\{ 1, \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)} \right\}.$$

Hence

$$\begin{aligned} \pi(x)P(x, y) &= \pi(x)Q(x, y) \min \left\{ 1, \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)} \right\} \\ &= \min\{\pi(x)Q(x, y), \pi(y)Q(y, x)\}. \end{aligned} \tag{23}$$

Interchanging the roles of x and y gives

$$\pi(y)P(y, x) = \min\{\pi(x)Q(x, y), \pi(y)Q(y, x)\}.$$

Therefore

$$\pi(x)P(x, y) = \pi(y)P(y, x),$$

so detailed balance holds.

The result now follows immediately from the theorem proved in the previous section.

□

8 Random Walk Metropolis

One of the simplest choices of proposal distribution is obtained by making a small random perturbation of the current state.

Suppose that the current state is

$$\theta^{(t)}.$$

Generate a random vector

$$Z$$

having a distribution that is symmetric about the origin and propose

$$Y = \theta^{(t)} + Z.$$

For example,

$$Z \sim N(0, \Sigma)$$

or

$$Z \sim \text{Uniform}([-a_1, a_1] \times \cdots \times [-a_d, a_d]).$$

The proposal density is therefore

$$Q(x, y) = f_Z(y - x), \tag{24}$$

where f_Z denotes the density of Z .

Since the distribution of Z is symmetric,

$$f_Z(z) = f_Z(-z), \tag{25}$$

which immediately implies

$$Q(x, y) = Q(y, x). \tag{26}$$

Thus the proposal mechanism is equally likely to move from x to y as from y back to x .

Substituting (26) into the Metropolis–Hastings acceptance probability gives

$$\begin{aligned} A(x, y) &= \min \left\{ 1, \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)} \right\} \\ &= \min \left\{ 1, \frac{\pi(y)}{\pi(x)} \right\}. \end{aligned} \tag{27}$$

This algorithm is known simply as the *Metropolis algorithm*.

Notice that the proposal distribution disappears completely from the acceptance probability. Its only role is to generate candidate moves.

The Metropolis algorithm therefore consists of the following two steps.

1. Generate a proposal

$$Y = X + Z.$$

2. Accept the proposal with probability

$$\min \left\{ 1, \frac{\pi(Y)}{\pi(X)} \right\}.$$

The interpretation is simple.

If the proposal moves toward a region where the target density is higher, then

$$\pi(Y) \geq \pi(X),$$

and the proposal is accepted with probability one.

If the proposal moves toward a region of lower probability, it is accepted only with probability

$$\frac{\pi(Y)}{\pi(X)}.$$

Thus the Markov chain tends to move toward regions where the target density is large while still occasionally exploring lower-density regions. These occasional downhill moves are crucial; without them the chain would become trapped at local modes.

8.1 Choice of Proposal Distribution

Although the proposal distribution does not appear explicitly in the acceptance probability, it has a profound effect on the efficiency of the algorithm.

Suppose the proposal variance is extremely small.

Then almost every proposal is accepted because the proposed point is very close to the current state.

However, the chain moves only by tiny steps and requires a very long time to explore the state space.

At the other extreme, suppose the proposal variance is extremely large.

The proposed values are now far away from the current state and often lie in regions where the target density is very small. Consequently, most proposals are rejected and the chain again moves very slowly.

Thus there is an important trade-off.

Small proposal variances lead to high acceptance probabilities but slow exploration.

Large proposal variances lead to rapid exploration but low acceptance probabilities.

An effective proposal distribution balances these two competing effects.

9 Example: Sampling from a Power-Law Distribution

Let

$$S = \{1, 2, \dots, M\},$$

and suppose our goal is to sample from the probability mass function

$$\pi(i) = \frac{i^{-\alpha}}{\sum_{k=1}^M k^{-\alpha}}, \quad \alpha > 0. \quad (28)$$

Distributions of this form arise in many applications including network science, linguistics, genetics and statistical physics.

Observe that the normalizing constant

$$\sum_{k=1}^M k^{-\alpha}$$

may be expensive to compute when the state space is large.

Fortunately, the Metropolis algorithm does not require this normalizing constant.

We use the following proposal mechanism.

If the current state is

$$i,$$

then

- propose $i + 1$ with probability $1/2$,
- propose $i - 1$ with probability $1/2$,

with the obvious modifications at the endpoints.

Since the proposal is symmetric,

$$Q(i, j) = Q(j, i),$$

the acceptance probability is

$$\begin{aligned} A(i, j) &= \min \left\{ 1, \frac{\pi(j)}{\pi(i)} \right\} \\ &= \min \left\{ 1, \left(\frac{i}{j} \right)^\alpha \right\}. \end{aligned} \quad (29)$$

Suppose, for example, that the chain is currently at state 20 and proposes to move to state 21.

The acceptance probability is

$$\left(\frac{20}{21}\right)^\alpha,$$

which is slightly smaller than one.

On the other hand, if the proposal is to move from 20 to 19, then

$$\left(\frac{20}{19}\right)^\alpha > 1,$$

so the move is accepted with probability one.

Thus the chain naturally tends to drift toward smaller integers, where the target probability is larger, while still occasionally moving toward larger integers.

Most importantly, notice that the normalizing constant in (28) never appears in the acceptance probability. Only the ratio of the unnormalized probabilities is required.

This simple example illustrates one of the principal strengths of the Metropolis algorithm: it remains applicable even when the target distribution is known only up to proportionality.

10 Why Metropolis–Hastings is So Powerful

The Metropolis–Hastings algorithm possesses several remarkable features which explain why it has become one of the most widely used algorithms in Bayesian statistics.

10.1 The Normalizing Constant Cancels

Suppose the target density can be written as

$$\pi(\theta) = \frac{\tilde{\pi}(\theta)}{Z},$$

where

$$Z = \int \tilde{\pi}(\theta) d\theta$$

is an unknown normalizing constant.

Substituting this expression into the acceptance probability gives

$$\frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)} = \frac{\tilde{\pi}(y)Q(y, x)}{\tilde{\pi}(x)Q(x, y)}. \quad (30)$$

The unknown constant Z disappears completely.

Consequently, Metropolis–Hastings requires only pointwise evaluation of an unnormalized density.

This is precisely the situation encountered in Bayesian statistics, where Bayes' theorem gives

$$\pi(\theta \mid \text{data}) \propto L(\theta)p(\theta).$$

The marginal likelihood never needs to be computed.

10.2 The Proposal Distribution is Flexible

The proposal distribution

$$Q(x, \cdot)$$

may be chosen almost arbitrarily.

It may be

- a random walk,
- a Gaussian distribution,
- a uniform distribution,
- a distribution depending on the current state,
- or any other distribution that is easy to simulate.

Different proposal distributions lead to different efficiencies, but they all target the same stationary distribution after the appropriate acceptance step is added.

10.3 Local Moves Produce Global Sampling

Notice that the algorithm never attempts to sample directly from the target distribution.

Instead, it repeatedly performs small local moves.

Although each individual move is simple, the resulting trajectory eventually explores the entire state space.

This idea is common throughout modern computational statistics.

Rather than solving a difficult problem in one step, we repeatedly perform many simple stochastic updates whose long-run behaviour is easy to analyse.

10.4 The Price We Pay

Nothing comes for free.

The samples generated by Metropolis–Hastings are not independent.

Indeed,

$$\theta^{(t+1)}$$

is generated from

$$\theta^{(t)},$$

so successive iterates are necessarily correlated.

This dependence reduces the amount of information contained in a fixed number of samples compared with independent Monte Carlo sampling.

Nevertheless, because independent sampling from the posterior is usually impossible, this is a very reasonable price to pay.

The dependence becomes less important as the chain mixes rapidly, which is why careful choice of the proposal distribution is important.

11 Gibbs Sampling

The Metropolis–Hastings algorithm is remarkably general. However, it has one potential drawback. Every proposed move may be rejected.

If the proposal distribution is poorly chosen, the Markov chain may remain at the same state for many consecutive iterations, causing the algorithm to explore the state space very slowly.

This naturally raises the following question.

Can we construct an MCMC algorithm in which every proposed move is accepted?

Surprisingly, the answer is yes.

The resulting algorithm is known as the *Gibbs sampler*. It is one of the most important algorithms in Bayesian computation.

11.1 The Basic Idea

Suppose the unknown parameter is

$$\theta = (\theta_1, \theta_2, \dots, \theta_p),$$

and suppose our target distribution is

$$\pi(\theta_1, \dots, \theta_p).$$

Assume that although sampling directly from the joint distribution is difficult, sampling from each full conditional distribution is easy.

More precisely, suppose we can generate samples from

$$\pi(\theta_1 \mid \theta_2, \dots, \theta_p),$$

$$\pi(\theta_2 \mid \theta_1, \theta_3, \dots, \theta_p),$$

and so on.

The idea of Gibbs sampling is extremely simple.

Rather than proposing an entirely new parameter vector, we update only one coordinate at a time by sampling exactly from its conditional distribution.

11.2 The Gibbs Sampling Algorithm

Suppose the current state of the Markov chain is

$$\theta^{(t)} = (\theta_1^{(t)}, \dots, \theta_p^{(t)}).$$

One iteration of the Gibbs sampler proceeds as follows.

1. Sample

$$\theta_1^{(t+1)} \sim \pi(\theta_1 \mid \theta_2^{(t)}, \dots, \theta_p^{(t)}).$$

2. Sample

$$\theta_2^{(t+1)} \sim \pi(\theta_2 \mid \theta_1^{(t+1)}, \theta_3^{(t)}, \dots, \theta_p^{(t)}).$$

3. Continue in this fashion until

$$\theta_p^{(t+1)} \sim \pi(\theta_p \mid \theta_1^{(t+1)}, \dots, \theta_{p-1}^{(t+1)}).$$

After completing these updates, one full Gibbs iteration has been performed.

Notice that newly updated coordinates are immediately used in later updates during the same iteration.

11.3 Two-Dimensional Gibbs Sampling

To illustrate the algorithm, suppose

$$\theta = (\theta_1, \theta_2).$$

The Gibbs sampler alternates between the two conditional distributions.

Starting from

$$(\theta_1^{(0)}, \theta_2^{(0)}),$$

repeat

1. Sample

$$\theta_1^{(t+1)} \sim \pi(\theta_1 \mid \theta_2^{(t)}).$$

2. Sample

$$\theta_2^{(t+1)} \sim \pi(\theta_2 \mid \theta_1^{(t+1)}).$$

Each update changes only one coordinate while keeping the other coordinate fixed.

Consequently, the Markov chain gradually explores the entire two-dimensional state space.

11.4 Why Gibbs Sampling is Attractive

Compared with the Metropolis algorithm, Gibbs sampling has several important advantages.

1. There is no tuning parameter such as a proposal variance.
2. Every update is accepted.
3. The algorithm is often straightforward once the conditional distributions have been derived.

For these reasons, Gibbs sampling became one of the most widely used algorithms in Bayesian statistics, particularly before the development of modern Hamiltonian Monte Carlo methods.

Its principal disadvantage is that it requires the conditional distributions to be available in a form from which exact simulation is possible.

In many models these conditional distributions are known explicitly. Such models are often called *conditionally conjugate* models.

When exact sampling from the conditional distributions is impossible, one usually replaces the Gibbs updates by Metropolis–Hastings updates, leading to algorithms known collectively as *Metropolis-within-Gibbs*.

12 Gibbs Sampling as a Special Case of Metropolis–Hastings

At first sight, Gibbs sampling and the Metropolis–Hastings algorithm appear to be completely different.

The Metropolis–Hastings algorithm proposes a move and then accepts or rejects it.

The Gibbs sampler simply replaces one coordinate by a draw from the corresponding conditional distribution and never rejects the proposed value.

Remarkably, the Gibbs sampler is actually a special case of the Metropolis–Hastings algorithm.

12.1 The Proposal Distribution

For simplicity, suppose that

$$\theta = (\theta_1, \theta_2),$$

although exactly the same argument works in any dimension.

Suppose the current state is

$$x = (x_1, x_2).$$

The Gibbs sampler updates only the first coordinate.

It generates

$$y_1 \sim \pi(\theta_1 \mid x_2),$$

and proposes the new state

$$y = (y_1, x_2).$$

Notice that this proposal is itself a random variable.

Thus the proposal density is

$$Q(x, y) = \pi(y_1 \mid x_2). \tag{31}$$

Now suppose that the Markov chain is currently at the proposed state

$$y = (y_1, x_2).$$

Running exactly the same Gibbs update would generate

$$x_1 \sim \pi(\theta_1 \mid x_2),$$

and therefore

$$Q(y, x) = \pi(x_1 \mid x_2). \tag{32}$$

12.2 The Metropolis–Hastings Ratio

Recall that the Metropolis–Hastings acceptance probability is

$$A(x, y) = \min \left\{ 1, \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)} \right\}. \quad (33)$$

Substituting (31) and (32) gives

$$\frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)} = \frac{\pi(y_1, x_2) \pi(x_1 | x_2)}{\pi(x_1, x_2) \pi(y_1 | x_2)}. \quad (34)$$

Now use the factorization

$$\pi(\theta_1, \theta_2) = \pi(\theta_1 | \theta_2) \pi(\theta_2).$$

Equation (34) becomes

$$\begin{aligned} \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)} &= \frac{\pi(y_1 | x_2) \pi(x_2) \pi(x_1 | x_2)}{\pi(x_1 | x_2) \pi(x_2) \pi(y_1 | x_2)} \\ &= 1. \end{aligned} \quad (35)$$

Therefore

$$A(x, y) = 1.$$

Every proposed Gibbs move is accepted.

Theorem 12.1. *The Gibbs sampler is a special case of the Metropolis–Hastings algorithm in which every proposal is accepted.*

The above calculation explains why Gibbs sampling is so efficient.

The proposal distribution is chosen to be exactly the conditional distribution of the target distribution.

Consequently, the proposal already satisfies the balance required by the Metropolis–Hastings algorithm, making any additional rejection step unnecessary.

12.3 Metropolis-within-Gibbs

The previous derivation suggests an important generalization.

Suppose that one of the conditional distributions cannot be sampled exactly.

For example, suppose that

$$\pi(\theta_1 \mid \theta_2)$$

is available only up to a normalizing constant.

Instead of drawing exactly from this conditional distribution, we may perform a Metropolis–Hastings update targeting this conditional distribution while keeping

$$\theta_2$$

fixed.

The remaining coordinates may still be updated using ordinary Gibbs steps.

Algorithms of this form are known as *Metropolis-within-Gibbs* algorithms.

They combine the simplicity of Gibbs sampling with the generality of Metropolis–Hastings and are widely used in practical Bayesian computation.

13 Practical Considerations

The Metropolis–Hastings algorithm and the Gibbs sampler produce a sequence

$$\theta^{(0)}, \theta^{(1)}, \theta^{(2)}, \dots$$

whose stationary distribution is the desired target distribution. Several practical issues arise when implementing these algorithms.

13.1 Burn-in

The Markov chain is usually started from an arbitrary initial value.

Consequently, the early iterations need not resemble samples from the target distribution.

Fortunately, provided the chain is irreducible and aperiodic, the distribution of

$$\theta^{(t)}$$

approaches the stationary distribution as t increases.

For this reason it is common practice to discard the first several iterations of the chain. This initial portion is called the *burn-in period*.

After burn-in, the remaining iterates are approximately distributed according to the target distribution.

13.2 Autocorrelation

Unlike ordinary Monte Carlo sampling, successive MCMC samples are not independent.

Indeed,

$$\theta^{(t+1)}$$

is generated directly from

$$\theta^{(t)},$$

so nearby iterates tend to be highly correlated.

Strong autocorrelation implies that the Markov chain explores the state space slowly. Consequently, a long MCMC run may contain much less information than the same number of independent samples.

The objective when designing an efficient MCMC algorithm is therefore to reduce this dependence as much as possible while preserving the correct stationary distribution.

13.3 Trace Plots

One of the simplest diagnostic tools is the trace plot.

A trace plot is obtained by plotting

$$\theta^{(t)}$$

against the iteration number t .

A good trace plot typically fluctuates rapidly around a stable region without exhibiting long-term trends.

On the other hand, if the trace plot shows long periods during which the chain hardly moves, then the Markov chain is mixing slowly.

Trace plots provide a simple visual diagnostic and are routinely used to assess the behaviour of MCMC algorithms.

13.4 Choice of Proposal Distribution

For the Metropolis algorithm, the proposal distribution plays an important role.

If the proposal variance is too small, almost every proposal is accepted. However, the chain moves only by tiny steps and explores the state space very slowly.

If the proposal variance is too large, most proposed values fall in regions where the target density is very small and are therefore rejected.

In this case the chain again moves very slowly because it spends much of its time remaining at the current state.

An effective proposal distribution balances these two competing effects.

13.5 Summary

Markov Chain Monte Carlo provides a general methodology for sampling from complicated probability distributions.

The central idea is to construct a Markov chain whose stationary distribution is the desired target distribution.

The Metropolis–Hastings algorithm accomplishes this by correcting an arbitrary proposal mechanism through an acceptance-rejection step so that detailed balance holds.

The Gibbs sampler is a particularly elegant special case in which the proposal distribution is chosen to be the corresponding conditional distribution. Consequently, every proposed move is accepted.

These algorithms form the foundation of modern Bayesian computation and have applications throughout statistics, machine learning, statistical physics, computational biology and many other fields.

14 The Gibbs Sampler Recap

Let us introduce some notation. The goal is to obtain samples from a target distribution with density π (in our applications, π will be the posterior density). π represents the density on $\mathbb{R}^d$ of a random vector Θ of size $d \times 1$:

$$\Theta \sim \pi.$$

We assume that Θ can be decomposed as:

$$\Theta = (\Theta_{(1)}, \dots, \Theta_{(k)})$$

for k subvectors $\Theta_{(1)}, \dots, \Theta_{(k)}$. The j^{th} subvector is $\Theta_{(j)}$. We shall also denote the vector obtained by dropping the j^{th} subvector $\Theta_{(j)}$ from Θ by $\Theta_{-(j)}$. For example,

$$\Theta_{-(1)} = (\Theta_{(2)}, \dots, \Theta_{(k)}) \quad \text{and} \quad \Theta_{-(2)} = (\Theta_{(1)}, \Theta_{(3)}, \dots, \Theta_{(k)}).$$

Gibbs sampling works according to the following algorithm.

1. Initialize with an arbitrary value $\theta^{(0)}$.
2. Repeat the following for $t = 1, \dots, N$ (for a large number N):
 - a) Pick J uniformly at random from $1, \dots, k$. Suppose J turned out to be j .
 - b) Set $\theta_{-(j)}^{(t+1)} = \theta_{-(j)}^{(t)}$ i.e., $\theta_{(i)}^{(t+1)} = \theta_{(i)}^{(t)}$ for $i \neq j$.
 - c) $\theta_{(j)}^{(t+1)}$ is randomly drawn according to the conditional distribution of $\Theta_{(j)}$ given $\Theta_{-(j)} = \theta_{-(j)}^{(t)}$.

This algorithm assumes that we have some way of generating observations from the conditional distribution of $\Theta_{(j)}$ given $\Theta_{-(j)}$ for each j .

15 Examples of the Gibbs Sampler

In the first two examples below, the posterior can be written in closed form so MCMC is not really necessary. These are still useful for evaluating the performance of the Gibbs sampler. The third example is more non-trivial and illustrates the usefulness of the Gibbs sampler.

15.1 Bivariate Normal

We want to sample (θ_1, θ_2) from the following bivariate distribution:

$$\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} \bigg| \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \sim N \left(\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right).$$

Here $y = (y_1, y_2)^T$ is fixed. One can, of course, directly sample from this normal distribution without any MCMC. We can use this example to illustrate the performance of the Gibbs sampler. To use the Gibbs sampler, observe that

$$\theta_1 | \theta_2, y \sim N(y_1 + \rho(\theta_2 - y_2), 1 - \rho^2) \quad \text{and} \quad \theta_2 | \theta_1, y \sim N(y_2 + \rho(\theta_1 - y_1), 1 - \rho^2)$$

The Gibbs sampler algorithm then would be the following:

1. Initialize at arbitrary $\theta_1^{(0)}$ and $\theta_2^{(0)}$.
2. Repeat the following for $t = 1, \dots, N$:
 - a) Pick J to be either 1 or 2 with probability 0.5.
 - b) If $J = 1$, take $\theta_2^{(t+1)} = \theta_2^{(t)}$ and generate $\theta_1^{(t+1)}$ from the normal distribution with mean $y_1 + \rho(\theta_2^{(t)} - y_2)$ and variance $1 - \rho^2$.
 - c) If $J = 2$, take $\theta_1^{(t+1)} = \theta_1^{(t)}$ and generate $\theta_2^{(t+1)}$ from the normal distribution with mean $y_2 + \rho(\theta_1^{(t)} - y_1)$ and variance $1 - \rho^2$.

This example is also useful for illustrating one potential problem with the Gibbs sampler. Suppose $\rho = 1$. In this case $\theta_1 - y_1 \sim N(0, 1)$ and $\theta_2 - y_2 = \theta_1 - y_1$. The Gibbs iterates in this case will be

$$\theta_1^{(t+1)} - y_1 = \theta_2^{(t)} - y_2 \quad \text{if } J = 1$$

and

$$\theta_2^{(t+1)} - y_2 = \theta_1^{(t)} - y_1 \quad \text{if } J = 2$$

This results in the chain not moving after the first iterate. For example, if the first step is:

$$\begin{pmatrix} \theta_1^{(0)} \\ \theta_2^{(0)} \end{pmatrix} \rightarrow \begin{pmatrix} y_1 - y_2 + \theta_2^{(0)} \\ \theta_2^{(0)} \end{pmatrix},$$

then the chain will stay at $\begin{pmatrix} y_1 - y_2 + \theta_2^{(0)} \\ \theta_2^{(0)} \end{pmatrix}$ forever. On the other hand, if the first step is

$$\begin{pmatrix} \theta_1^{(0)} \\ \theta_2^{(0)} \end{pmatrix} \rightarrow \begin{pmatrix} \theta_1^{(0)} \\ y_2 - y_1 + \theta_1^{(0)} \end{pmatrix},$$

then the chain will stay forever at $\begin{pmatrix} \theta_1^{(0)} \\ y_2 - y_1 + \theta_1^{(0)} \end{pmatrix}$. This chain will obviously not converge unless the initial iterate already has the right distribution. A similar problem arises when $\rho = -1$.

The lesson to be drawn from this is that the Gibbs sampler has issues of convergence when there is high correlation between the variables.

15.2 Linear Regression

Consider the linear regression model that we discussed previously. We observe data $(y_1, x_1), \dots, (y_n, x_n)$ with $y_1, \dots, y_n$ real-valued and $x_1, \dots, x_n \in \mathbb{R}^p$. We treat $x_1, \dots, x_n$ as non-random and $y_1, \dots, y_n$ as random. The parameter is $\theta \in \mathbb{R}^{p+1}$ with $\theta = (\beta, \sigma)$ (here $\beta \in \mathbb{R}^p$ represents the vector of regression coefficients and σ is the noise standard deviation). Consider the model

$$y_i \sim N(x_i' \beta, \sigma^2)$$

with $y_1, \dots, y_n$ being independent conditional on θ along with the (improper) prior:

$$\pi(\beta, \sigma) \propto \frac{1}{\sigma} I\{\sigma > 0\}.$$

The exact joint posterior is given by

$$\pi(\beta, \sigma | \text{data}) \propto \sigma^{-(n+1)} \exp\left(\frac{-1}{2\sigma^2} \|Y - X\beta\|^2\right) I\{\sigma > 0\} \quad (36)$$

We can directly sample from this posterior (e.g., by first sampling from the marginal posterior of σ and then from the conditional posterior of β given σ). Alternatively, one can first sample from the marginal posterior of β and then from the conditional posterior of σ given β .

Now let us use the Gibbs sample to simulate from (36). Note that:

$$\pi(\beta | \sigma, \text{data}) \propto \exp\left(\frac{-1}{2\sigma^2} \|Y - X\beta\|^2\right) \propto \exp\left(\frac{-1}{2\sigma^2} \|X\beta - X\hat{\beta}\|^2\right) = \exp\left(\frac{-1}{2\sigma^2} (\beta - \hat{\beta})'(X'X)(\beta - \hat{\beta})\right)$$

which gives

$$\beta | \sigma, \text{data} \sim N_p\left(\hat{\beta}, \sigma^2(X'X)^{-1}\right).$$

Here $\hat{\beta}$ is the least squares estimator. For the conditional posterior of σ given β , note that

$$\pi(\sigma | \beta, \text{data}) \propto \sigma^{-(n+1)} \exp\left(\frac{-1}{2\sigma^2} \|Y - X\beta\|^2\right) I\{\sigma > 0\}.$$

Using the formula $f_{1/\sigma^2}(x) = f_\sigma(x^{-1/2})x^{-3/2}$, it is now easy to check that the density of $1/\sigma^2$ conditional on β and the data is proportional to

$$x^{(n-2)/2} \exp\left(\frac{-x}{2} \|Y - X\beta\|^2\right) I\{x > 0\}$$

which means that

$$\frac{1}{\sigma^2} \Big| \beta, \text{data} \sim \text{Gamma}\left(\frac{n}{2}, \frac{1}{2} \|Y - X\beta\|^2\right).$$

The Gibbs sampler algorithm is thus:

1. Initialize at arbitrary $\beta^{(0)}$ and $\sigma^{(0)}$.

2. Repeat the following for $t = 1, \dots, N$:

- a) Pick J to be either 1 or 2 with equal probability.
- b) If $J = 1$, take $\beta^{(t+1)} \sim N_p(\hat{\beta}, (\sigma^{(t)})^2 (X'X)^{-1})$ and $\sigma^{(t+1)} = \sigma^{(t)}$.
- c) If $J = 2$, take $\beta^{(t+1)} = \beta^{(t)}$. Generate an observation x from the Gamma distribution with parameters $n/2$ and $\|Y - X\beta^{(t)}\|^2/2$. Then take $\sigma^{(t+1)} = x^{-1/2}$.

15.3 Probit Regression

Our next application of the Gibbs sampler is to Probit Regression which was introduced in Homework Four. The model is given by the following. We observe data $(y_1, x_1), \dots, (y_n, x_n)$ with $y_1, \dots, y_n$ binary and $x_1, \dots, x_n \in \mathbb{R}^p$. $x_1, \dots, x_n$ are treated non-random and $y_1, \dots, y_n$ random. The parameter is $\beta \in \mathbb{R}^p$. The probit regression model assumes that $y_1, \dots, y_n$ are independent given β with

$$y_i | \beta \sim \text{Bernoulli}(\Phi(x_i' \beta))$$

where $\Phi(\cdot)$ is the standard normal cdf. Let $\pi(\cdot)$ be our prior for β . The posterior is then given by:

$$\pi(\beta | \text{data}) \propto \pi(\beta) \prod_{i=1}^n (\Phi(x_i' \beta))^{y_i} (1 - \Phi(x_i' \beta))^{1-y_i}.$$

Suppose we take the (improper) uniform prior for β so that the posterior becomes

$$\pi(\beta | \text{data}) \propto \prod_{i=1}^n (\Phi(x_i' \beta))^{y_i} (1 - \Phi(x_i' \beta))^{1-y_i}.$$

It is not actually clear how to use the Gibbs sampler on the above posterior. We can try decomposing β as $(\beta_1, \dots, \beta_p)$ and then attempt to obtain samples from the conditional:

$$\pi(\beta_j | \beta_{-j}, \text{data}) \propto \prod_{i=1}^n (\Phi(x_i' \beta))^{y_i} (1 - \Phi(x_i' \beta))^{1-y_i}.$$

I am not sure how to obtain samples from the above conditional.

There is a trick that people use to run the Gibbs sampler for probit regression. Let us introduce random variables $w_1, \dots, w_n$ which, conditional on β , are independent with

$$w_i | \beta \sim N(x_i' \beta, 1).$$

We then let

$$y_i = I\{w_i > 0\}.$$

It is then clear that $y_1, \dots, y_n$ defined thus have the same likelihood as in the probit regression model. Let $w = (w_1, \dots, w_n)$. The trick is to use the Gibbs sampler with (w, β) . We shall see how this is done in the next class.

16 Gibbs Sampler for Probit Regression

Our next application of the Gibbs sampler is to Probit Regression which was introduced in Homework Four. The model is given by the following. We observe data $(y_1, x_1), \dots, (y_n, x_n)$ with $y_1, \dots, y_n$ binary and $x_1, \dots, x_n \in \mathbb{R}^p$. $x_1, \dots, x_n$ are treated non-random and $y_1, \dots, y_n$

random. The parameter is $\beta \in \mathbb{R}^p$. The probit regression model assumes that $y_1, \dots, y_n$ are independent given β with

$$y_i | \beta \sim \text{Bernoulli}(\Phi(x'_i \beta))$$

where $\Phi(\cdot)$ is the standard normal cdf. Let $\pi(\cdot)$ be our prior for β . The posterior is then given by:

$$\pi(\beta | \text{data}) \propto \pi(\beta) \prod_{i=1}^n (\Phi(x'_i \beta))^{y_i} (1 - \Phi(x'_i \beta))^{1-y_i}.$$

Suppose we take the (improper) uniform prior for β so that the posterior becomes

$$\pi(\beta | \text{data}) \propto \prod_{i=1}^n (\Phi(x'_i \beta))^{y_i} (1 - \Phi(x'_i \beta))^{1-y_i}.$$

It is not actually clear how to use the Gibbs sampler on the above posterior. We can try decomposing β as $(\beta_1, \dots, \beta_p)$ and then attempt to obtain samples from the conditional:

$$\pi(\beta_j | \beta_{-j}, \text{data}) \propto \prod_{i=1}^n (\Phi(x'_i \beta))^{y_i} (1 - \Phi(x'_i \beta))^{1-y_i}.$$

I am not sure how to obtain samples from the above conditional.

There is a trick that people use to run the Gibbs sampler for probit regression. Let us introduce random variables $w_1, \dots, w_n$ which, conditional on β , are independent with

$$w_i | \beta \sim N(x'_i \beta, 1).$$

We then let

$$y_i = I\{w_i > 0\}.$$

It is then clear that $y_1, \dots, y_n$ defined thus have the same likelihood as in the probit regression model. Let $w = (w_1, \dots, w_n)$. The posterior of (w, β) given the data ($\text{data} = y = (y_1, \dots, y_n)$) is

$$\begin{aligned} \pi(w, \beta | \text{data}) &\propto \pi(\beta) \pi(w | \beta) \pi(y | w, \beta) \\ &\propto \prod_{i=1}^n \exp\left(-\frac{1}{2}(w_i - x'_i \beta)^2\right) \prod_{i=1}^n (I\{w_i > 0, y_i = 1\} + I\{w_i \leq 0, y_i = 0\}) \\ &\propto \exp\left(-\frac{1}{2}\|w - X\beta\|^2\right) \prod_{i=1}^n (I\{w_i > 0, y_i = 1\} + I\{w_i \leq 0, y_i = 0\}). \end{aligned}$$

where X is, as usual, the design matrix with rows $x'_1, \dots, x'_n$.

We shall now run the Gibbs sampler with the decomposition $\theta = (w, \beta)$. For this, we need to sample from the conditionals $\beta | w, \text{data}$ and $w | \beta, \text{data}$.

$$\begin{aligned} \pi(\beta | w, \text{data}) &\propto \pi(w, \beta | \text{data}) \\ &\propto \exp\left(-\frac{1}{2}\|w - X\beta\|^2\right) \prod_{i=1}^n (I\{w_i > 0, y_i = 1\} + I\{w_i \leq 0, y_i = 0\}) \\ &\propto \exp\left(-\frac{1}{2}\|w - X\beta\|^2\right) \end{aligned}$$

For a fixed $w \in \mathbb{R}^n$, let

$$\hat{\beta}_w := \underset{\beta \in \mathbb{R}^p}{\text{argmin}} \|w - X\beta\|^2 = (X'X)^{-1}X'w$$

so that

$$\|w - X\beta\|^2 = \|w - X\hat{\beta}_w\|^2 + \|X\hat{\beta}_w - X\beta\|^2 = \|w - X\hat{\beta}_w\|^2 + (\beta - \hat{\beta}_w)'(X'X)(\beta - \hat{\beta}_w).$$

Thus

$$\pi(\beta|w, \text{data}) \propto \exp\left(\frac{-1}{2}(\beta - \hat{\beta}_w)'(X'X)(\beta - \hat{\beta}_w)\right).$$

In other words,

$$\beta|w, \text{data} \sim N(\hat{\beta}_w, (X'X)^{-1}).$$

The next task is to figure out the conditional $w|\beta, \text{data}$:

$$\begin{aligned} \pi(w|\beta, \text{data}) &\propto \pi(w, \beta|\text{data}) \\ &\propto \exp\left(\frac{-1}{2}\|w - X\beta\|^2\right) \prod_{i=1}^n (I\{w_i > 0, y_i = 1\} + I\{w_i \leq 0, y_i = 0\}) \\ &\propto \prod_{i=1}^n \exp\left(\frac{-1}{2}(w_i - x'_i\beta)^2\right) (I\{w_i > 0, y_i = 1\} + I\{w_i \leq 0, y_i = 0\}) \end{aligned}$$

Thus $w_1, \dots, w_n$ are independent conditional on β and data. Further

$$\pi(w_i|\beta, \text{data}) = \pi(w_i|\beta, y_i) \propto \exp\left(\frac{-1}{2}(w_i - x'_i\beta)^2\right) (I\{w_i > 0, y_i = 1\} + I\{w_i \leq 0, y_i = 0\}).$$

It makes sense now to separate the two cases $y_i = 1$ and $y_i = 0$. For $y_i = 1$, we get

$$\pi(w_i|\beta, y_i = 1) \propto \exp\left(\frac{-1}{2}(w_i - x'_i\beta)^2\right) I\{w_i > 0\}.$$

Thus, conditional on β and $y_i = 1$, the random variable w_i has the $N(x'_i\beta, 1)$ distribution truncated to the positive half-line.

Here is a fact (from the wikipedia article for truncated normal distributions: https://en.wikipedia.org/wiki/Truncated_normal_distribution#Generating_values_from_the_truncated_normal_distribution) that is useful for simulating from truncated normal distributions.

Fact 16.1. *Consider a random variable X_{trunc} having the truncated $N(\mu, \sigma^2)$ truncated to the interval (a, b) (here $a < b$; a is allowed to be $-\infty$ and b is allowed to be $+\infty$). The density of X_{trunc} is equal to the conditional density of X conditioned on $a < X < b$ where $X \sim N(\mu, \sigma^2)$.*

X_{trunc} can be simulated using the equation:

$$X_{trunc} = \mu + \sigma\Phi^{-1}((1 - U)\Phi(a_0) + U\Phi(b_0)) \quad \text{where } a_0 := \frac{a - \mu}{\sigma} \text{ and } b_0 := \frac{b - \mu}{\sigma}$$

and U has the uniform distribution on $(0, 1)$. I will leave as an exercise for you to verify that the right hand side above indeed has the required truncated normal distribution.

Using this fact, it is straightforward to verify that $w_i|\{\beta, y_i = 1\}$ can be simulated by

$$x'_i\beta + \Phi^{-1}((1 - U)\Phi(-x'_i\beta) + U).$$

This follows directly from the fact above by taking $\mu = x'_i\beta$, $\sigma = 1$, $a = 0$ and $b = \infty$ (this implies that $a_0 = -x'_i\beta$ and $b_0 = \infty$).

One can argue analogously to deduce that $w_i|\{\beta, y_i = 0\}$ can be simulated by

$$x'_i\beta + \Phi^{-1}\left(U\Phi(-x'_i\beta)\right).$$

The Gibbs sampler for the probit model can be therefore implemented by the following model:

1. Initialize at $\beta^{(0)}$ and $w^{(0)}$.
2. Repeat the following for $t = 1, \dots, N$
 - a) Pick J to be either 1 or 2 with equal probability.
 - b) If $J = 1$, take $\beta^{(t+1)} \sim N((X'X)^{-1}X'w^{(t)}, (X'X)^{-1})$ and $w^{(t+1)} = w^{(t)}$.
 - c) If $J = 2$, take $\beta^{(t+1)} = \beta^{(t)}$ and

$$w_i = \begin{cases} x'_i\beta^{(t)} + \Phi^{-1}\left((1 - U_i)\Phi(-x'_i\beta^{(t)}) + U_i\right) & \text{if } y_i = 1 \\ x'_i\beta^{(t)} + \Phi^{-1}\left(U_i\Phi(-x'_i\beta^{(t)})\right) & \text{if } y_i = 0 \end{cases}$$

where $U_i \sim Unif(0, 1)$.